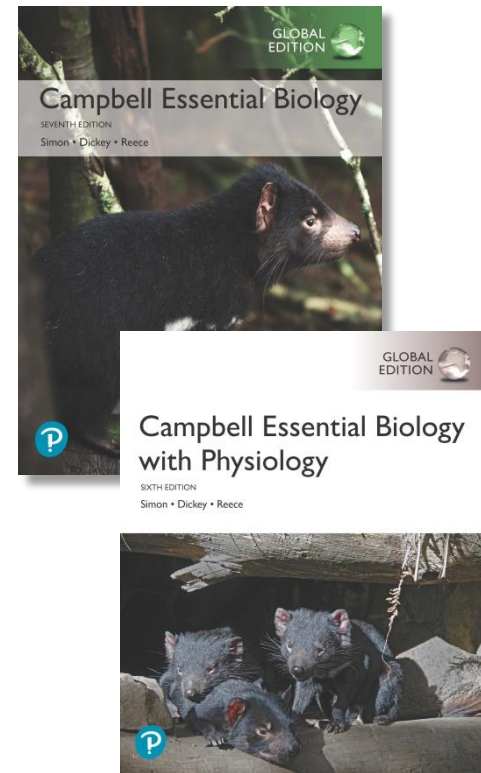


Campbell Essential Biology, Seventh Edition, Global Edition and Campbell Essential Biology with Physiology, Sixth Edition, Global Edition

Chapter 26 Reproduction and Development



PowerPoint® Lectures created by Edward J. Zalisko, Eric J. Simon, Jean L. Dickey, and Jane B. Reece

Nine Months May Seem Like A Long Time, But Elephant Pregnancy Drags On For 22 Months

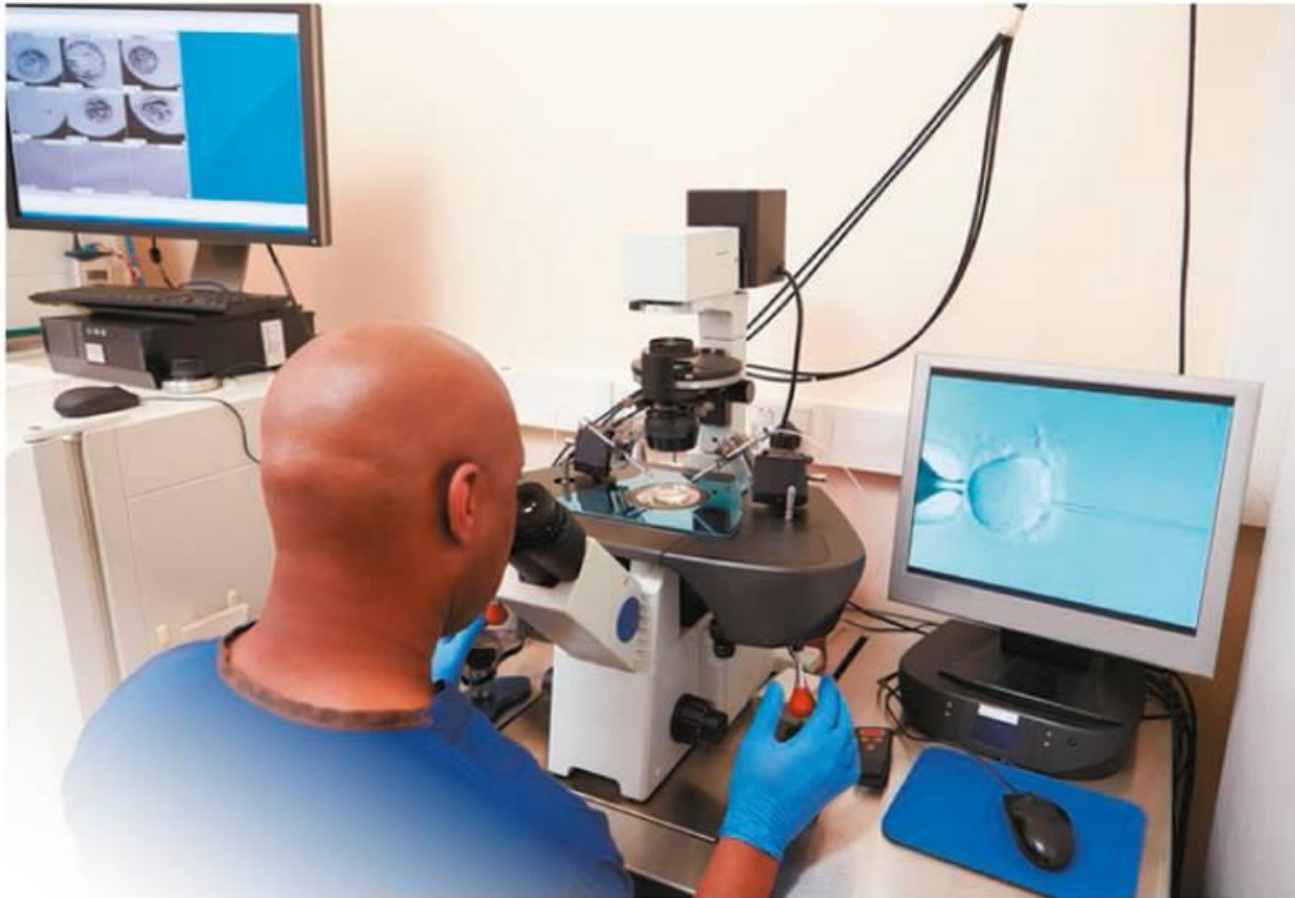


Biology and Society: New Ways of Making Babies (1 of 2)

- An infertile couple in New Jersey were using reproductive technologies to become pregnant.
 - But during those efforts, the husband suddenly died of a heart attack.
 - Within six hours of the death of her husband, the woman requested that a sample of her late husband's sperm be removed.
 - That sperm was used to fertilize the wife's eggs and produce a child who is alive today.
 - However, laws remain unclear regarding the rights of the child to death benefits.

Death benefits: 사망보험금

High-tech fertilization



Biology and Society: New Ways of Making Babies (2 of 2)

- This case illustrates two important points.
 - Modern technologies allow us to circumvent the natural course of reproduction in many ways.
 - Society's laws have not yet caught up to these new technologies.
 - It is up to all of us to decide what is right in such circumstances.
 - To quote the U.S. Court of Appeals in Philadelphia from a related case, “[We] cannot help but observe that this is, indeed, a new world.”

U.S. Court of Appeals: 미국연방 항소법원

Unifying Concepts of Animal Reproduction: Asexual Reproduction

- **Reproduction** is the **creation of new individuals from existing ones**.
- **Asexual reproduction** is reproduction without sex and creates offspring that are genetically identical to the parent.
 - **Budding** is the outgrowth and eventual splitting off of a new individual from the parent.
 - **Fission** is the separation of a parent into two or more offspring of about equal size.
 - **Fragmentation** is the breaking of the body into several pieces, followed by regeneration.

Hydra reproducing by budding



Sea anemone reproducing by **fission**



Asexual Reproduction

- Asexual reproduction
 - eliminates the need to find a mate,
 - allows the individual members of a species to perpetuate themselves even if they are isolated from one another, and
 - permits organisms to multiply quickly without spending time or energy producing sperm and eggs.
- But if the environment changes and becomes less favorable to survival, then all individuals may be affected equally, and the population may die out.

Sexual Reproduction (1 of 2)

- **Sexual reproduction** involves the **fusion of two haploid sex cells called gametes** (**sperm** and **egg**) and the formation of a **zygote**.
 - Sexual reproduction increases the genetic variability among offspring. Variation is the raw material of evolution by natural selection.
 - Some animals can reproduce sexually and asexually, benefiting from both modes.

Asexual (left) and sexual (right) reproduction in a starlet sea anemone



Sea anemone:
바다 말미잘

Sexual Reproduction (2 of 2)

- In some species, each individual is a **hermaphrodite**, with male and female reproductive systems. Some hermaphrodites can fertilize their own eggs, while other species require a partner.
- The mechanics of fertilization play an important part in sexual reproduction.
 - Many aquatic animals discharge their gametes into the water, where external fertilization occurs.
 - Nearly all terrestrial animals reproduce by internal fertilization, in which a male deposits sperm within a female's body.

Human Reproduction: Male Reproductive Anatomy

- Both sexes of humans have a pair of **gonads** that produce gametes, **ducts** to store and deliver the gametes, and **structures that allow mating**.
- **Figure 26.5** shows the male reproductive system.
 - The **penis** transfers sperm from the male to the female during sexual intercourse.
 - It contains three cylinders of erectile tissue that fill with blood and cause an erection during arousal.
 - The penis consists of a shaft that supports a highly sensitive glans. A **prepuce**, or foreskin, covers the glans, and may be removed by circumcision.

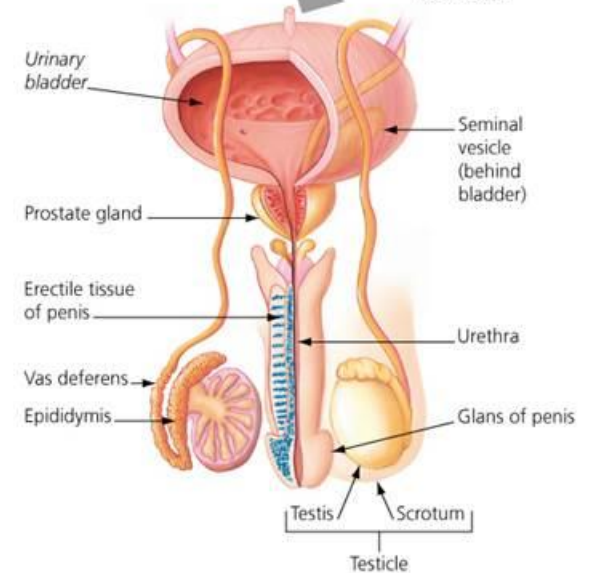
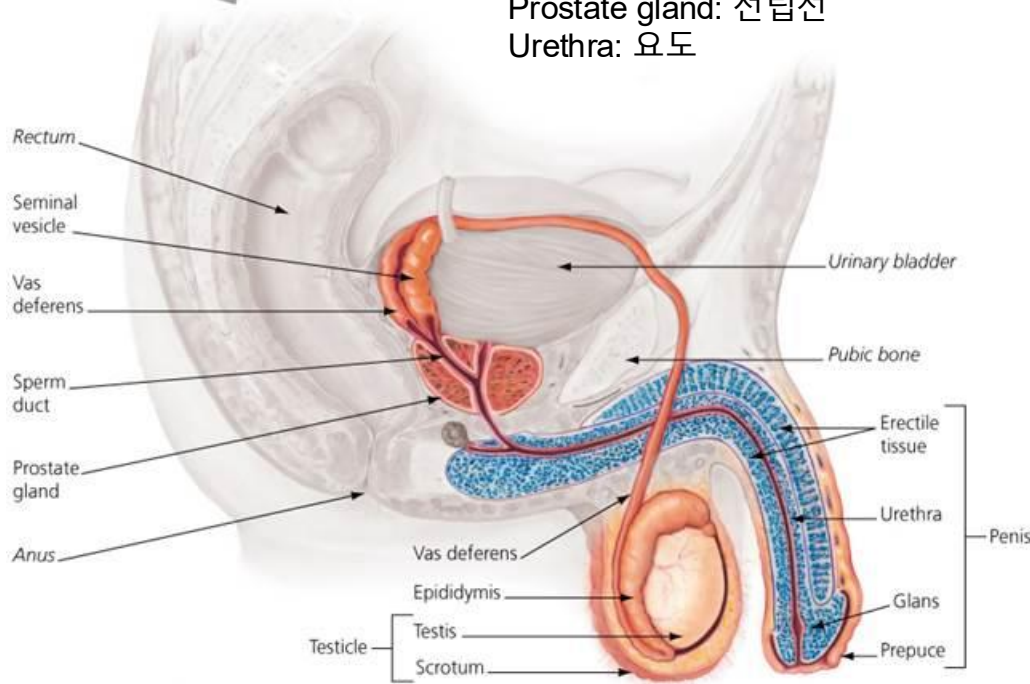
Glans: 귀두

Prepuce: 포피

The male reproductive system



Testes: 고환
 Scrotum: 음낭
 Testicle: 고환
 Epididymis: 부고환
 Vas deference: 정관
 Seminal vesicle: 정액주머니
 Sperm duct: 정액수송관
 Prostate gland: 전립선
 Urethra: 요도



Male Reproductive Anatomy (1 of 2)

- The male gonads, the **testes**, are located outside the abdominal cavity, are enclosed in a sac called the **scrotum**, and produce sperm.
 - A testis and scrotum together are called a **testicle**.
 - Because sperm do not develop optimally at body temperature, the scrotum promotes sperm formation by keeping sperm-forming cells about **2°C cooler**.
 - From puberty into old age, the testes produce hundreds of millions of sperm each day.
 - From the testes, sperm pass into a coiled tube called the **epididymis**.

Testes: 고환

Scrotum: 음낭

Testicle: 고환

Epididymis: 부고환

Male Reproductive Anatomy (2 of 2)

- During **ejaculation**, sperm leave the epididymis and travel through a duct called the **vas deferens**. The seminal vesicles and prostate gland add fluid that nourishes the sperm and provides protection from the natural acidity of the vagina.
 - Each vas deferens, one from each of the two testes, empties into the **urethra**, which at different times, transports sperm and urine out through the penis.
 - Ejaculation, caused by the contraction of muscles along the sperm ducts, releases about 5 mL of **semen**. Only 5% of semen consists of sperm (typically 200–500 million of them); the remaining 95% of semen is fluid secreted by the various glands.

Female Reproductive Anatomy (1 of 5)

- The outer features of the female reproductive tract are collectively called the vulva.
 - The **vagina**, or birth canal, opens to the outside just behind the opening of the urethra, the tube through which urine is excreted.
 - An outer pair of thick, fatty ridges, the labia majora, protects the entire genital region.
 - A pair of inner skin folds, the labia minora, borders the openings.

Vulva: 음문

Vagina: 질

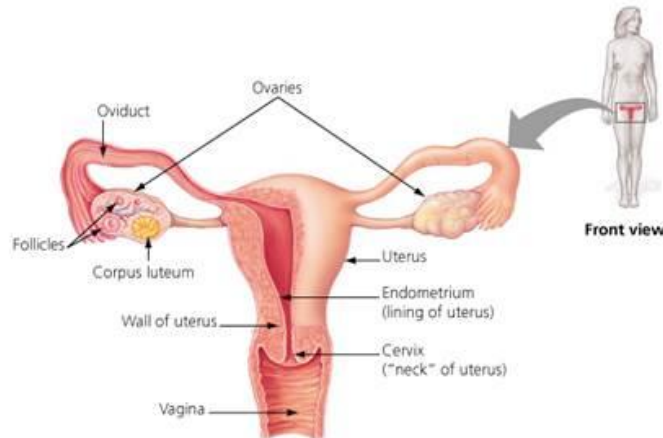
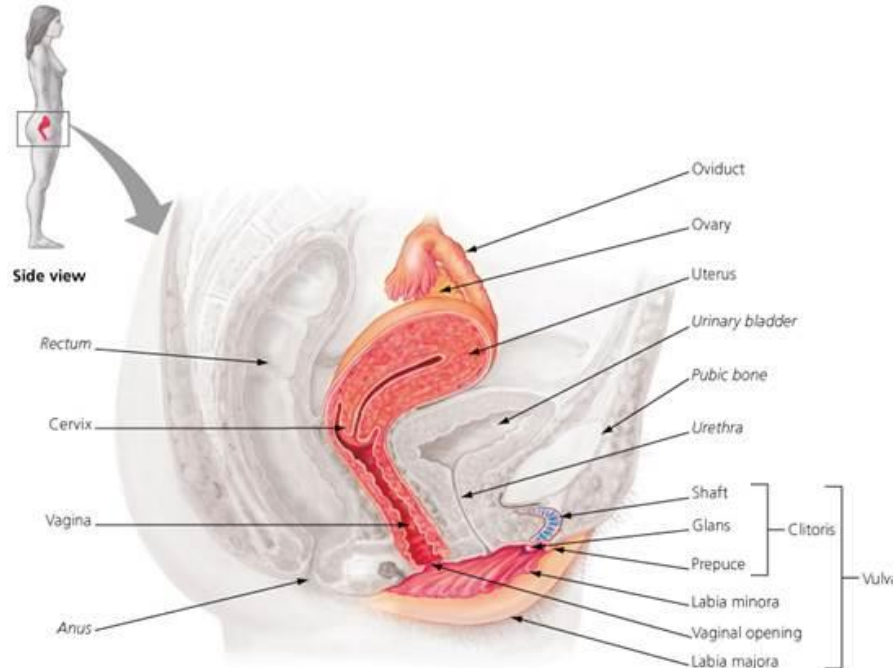
Labia minora: 대음순

Female Reproductive Anatomy (2 of 5)

- A thin membrane called the **hymen** partly covers the vaginal opening until sexual intercourse or other vigorous physical activity ruptures it.
- The clitoris is an organ that engorges with blood and becomes erect during sexual arousal.
 - The clitoris consists of a short shaft supporting a rounded glans, or head, covered by a small hood of skin called the prepuce.
 - The clitoris has an enormous number of nerve endings and is very sensitive to touch.

Hymen: 처녀막

The female reproductive system



Vulva: 음문
 Vagina: 질
 Labia minora: 대음순
 Cervix: 자궁경관
 Uterus: 자궁
 Ovary: 난소
 Oviduct: 난관
 Corpus luteum: 황체
 Endometrium: 자궁 내막

Female Reproductive Anatomy (3 of 5)

- The **ovaries** are the female gonads, the sites of gamete production.
 - Each ovary is about an inch long and has a bumpy surface. The bumps are follicles, each consisting of a single developing egg cell surrounded by cells that nourish and protect it. The follicles also produce estrogen, the female sex hormone.
 - Starting at puberty and continuing until menopause (the end of fertility, which usually occurs during middle age), a woman undergoes ovulation about every 28 days.

Female Reproductive Anatomy (4 of 5)

- During **ovulation**, an immature egg is ejected from a follicle and enters the **oviduct**, where fertilization typically occurs.
- The remains of the follicle grow to form a solid mass, the **corpus luteum**, which secretes hormones during the reproductive cycle.
- The **uterus** is the site of pregnancy. It is lined by a thick blood-rich layer of tissue, the **endometrium**, where an embryo implants and grows.

Ovulation: 배란

Female Reproductive Anatomy (5 of 5)

- A developing human is called an **embryo** for the first nine weeks (until body structures begin to appear) and **a fetus** from the 9th week until birth.
- The **cervix** is the narrow neck at the bottom of the pear-shaped uterus and opens into the vagina.
- Yearly Pap smears sample cervical cells and can detect cervical cancer early, when treatment is more likely to be successful.

Embryo: 배아

Fetus: 태아

Pap smear exam: 자궁경부암 진단검사

Gametogenesis

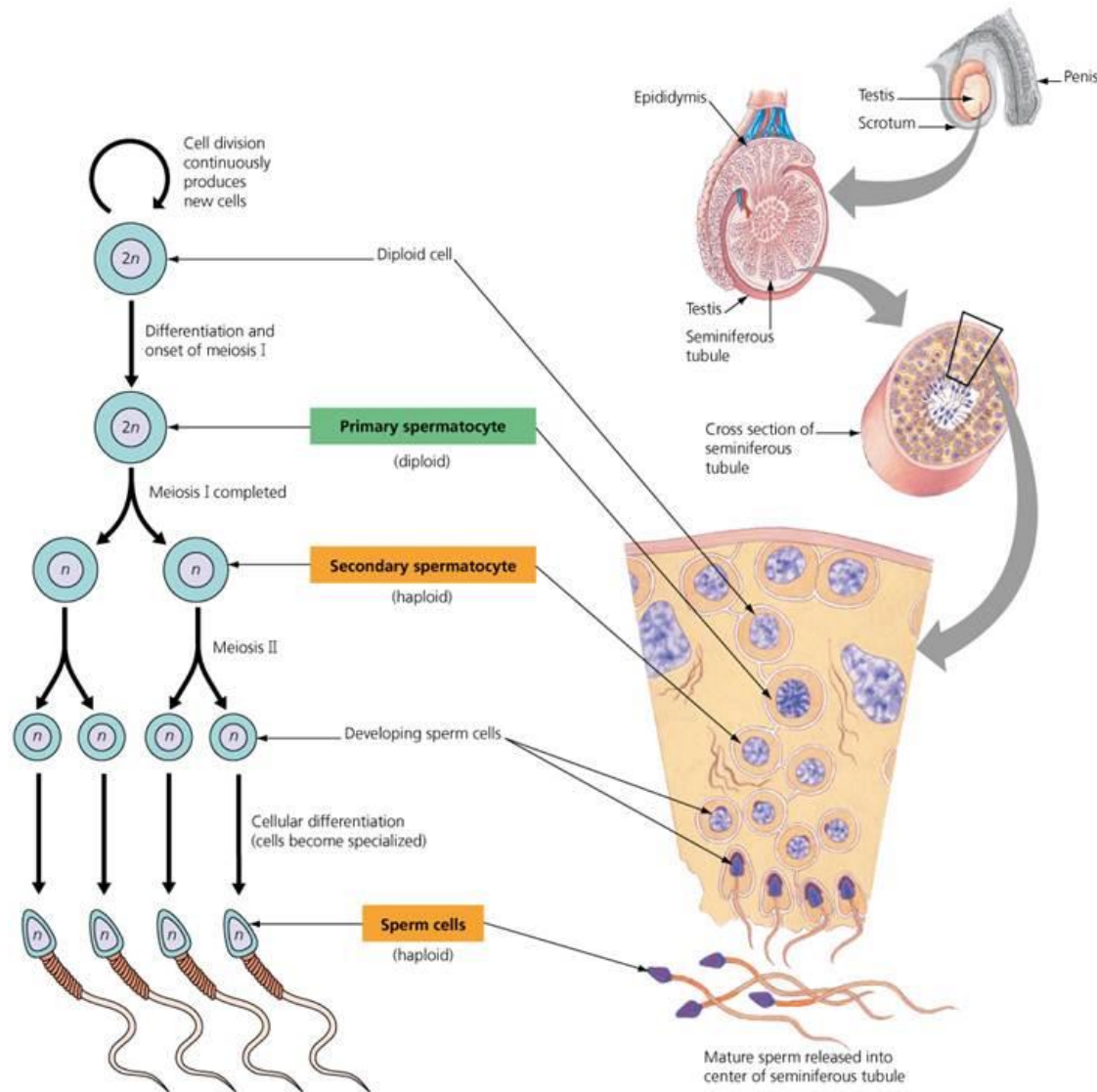
- The production of gametes is **gametogenesis**. Human gametes—sperm and egg—are haploid cells with 23 chromosomes that develop by meiosis from diploid cells with 46 chromosomes.

Spermatogenesis (1 of 2)

- **Spermatogenesis** is the formation of sperm cells.
 - Sperm cells develop inside the testes in coiled tubes called the **seminiferous tubules**.
 - **Primary spermatocytes** are diploid cells, undergo meiosis I, and produce **secondary spermatocytes**.
 - Secondary spermatocytes undergo meiosis II, each with the haploid number of chromosomes.
 - A sperm cell that develops from one of these haploid cells is gradually pushed toward the center of the seminiferous tubule. From there, it passes into the epididymis, where it matures, becomes motile, and is stored until ejaculation.

Seminiferous tubule: 정세관

Spermatogenesis (2 of 2)

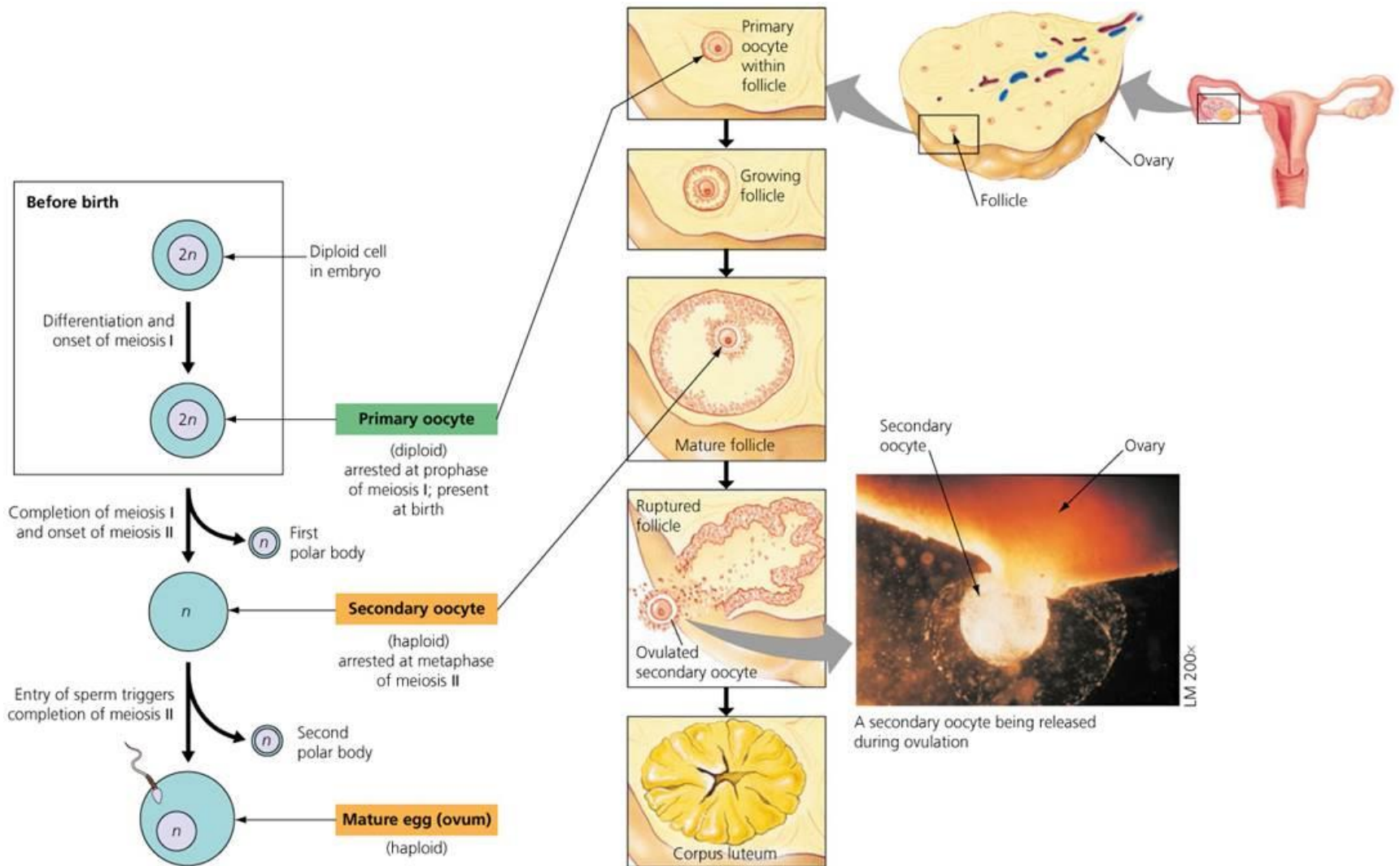


Oogenesis (1 of 2)

- **Oogenesis** is the development of egg cells.
- A primary oocyte
 - is a diploid cell that has paused its cell cycle in prophase of meiosis I and
 - can be triggered to develop further by the hormone FSH to produce a **secondary oocyte** and **a small polar body** with almost no cytoplasm.
 - Meiosis II is also unequal, yielding **a small polar body and a mature egg**.

Oogenesis: 난자 형성

Oogenesis (2 of 2)



Gamete Formation in Males versus Females

- Both oogenesis and spermatogenesis produce haploid gametes, but there are **several important differences** between the two processes.
 - Human males create new sperm **every day**, from puberty through old age. Human females **create primary oocytes only during fetal development**.
 - From each parent cell spermatogenesis produces **four** gametes but oogenesis produces **only one**.
 - Sperm are small, move by means of a whip-like flagellum, and contain relatively few nutrients. Eggs are large, not self-propelled, and well stocked with nutrients and organelles.

The Female Reproductive Cycle (1 of 3)

- Human females have a **reproductive cycle**, a recurring series of events that produces gametes, makes eggs available for fertilization, and prepares the body for pregnancy.
 - The reproductive cycle repeats every 28 days, on average, but cycles from 20 to 40 days are not uncommon.
 - The reproductive cycle is actually two cycles in one.
 - **The ovarian cycle** controls the growth and release of an egg. During **the menstrual cycle** the uterus is prepared for possible implantation of an embryo.

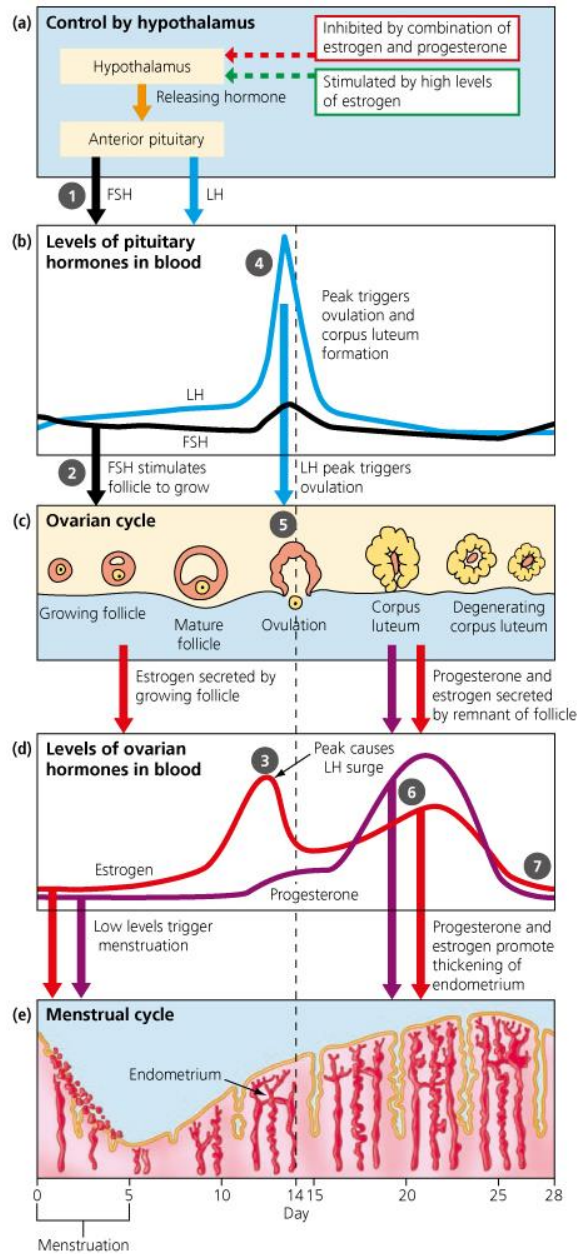
The Female Reproductive Cycle (2 of 3)

- Hormonal messages coordinate the two cycles, keeping them synchronized with each other.
- The reproductive cycle is a good example of the importance of interactions—in this case, among the nervous, endocrine, and reproductive systems—that control bodily functions.
- **Menstruation** is uterine bleeding caused by the breakdown of the endometrium, the blood-rich inner lining of the uterus. If an embryo implants in the uterine wall, it will obtain nutrients from the endometrium, and the thickened lining will not be discharged. Menstruation is thus a sign that pregnancy has not occurred in the previous cycle.

The Female Reproductive Cycle (3 of 3)

- The hormones shown in **Figure 26.9** regulate the ovarian and menstrual cycles, synchronizing ovulation with preparation of the uterus for possible implantation of an embryo.
- If an embryo has implanted in the endometrium, it secretes a hormone called HCG (human chorionic gonadotropin). HCG maintains the corpus luteum, which continues to secrete progesterone and estrogen, keeping the endometrium intact.

The reproductive cycle of the human female

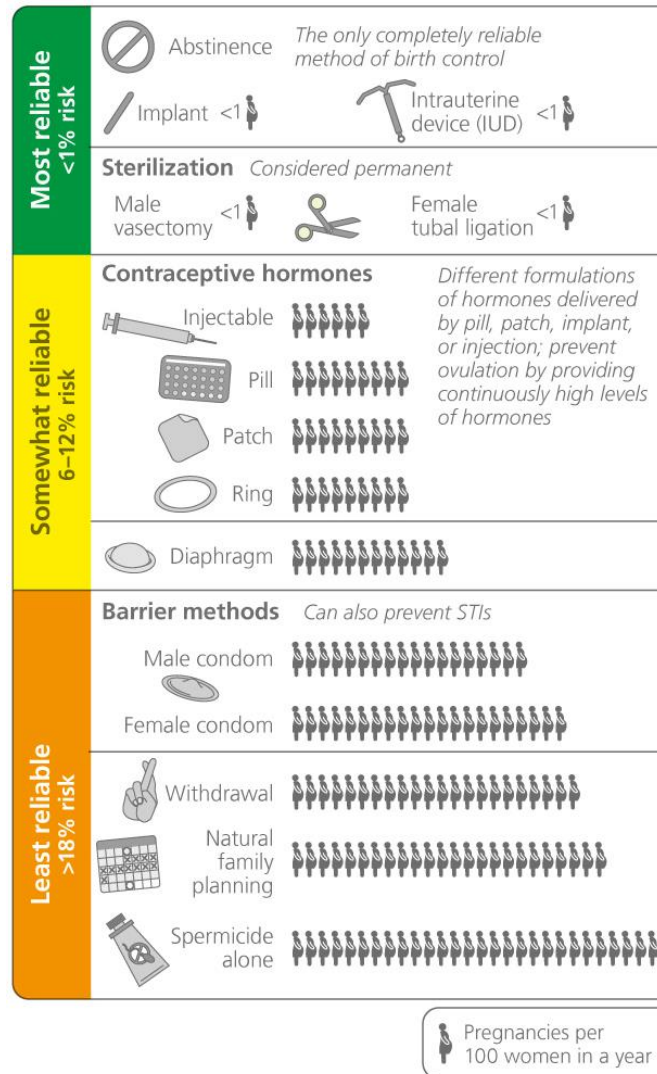


Follicle stimulating hormone: 여포 자극 호르몬
Luteinizing hormone: 황체 형성 호르몬

Reproductive Health: Contraception

- **Contraception** is the deliberate prevention of pregnancy.
 - There are many forms of contraception that interfere with different steps in the process of becoming pregnant.
 - **Figure 26.10** lists different methods of birth control sorted by their failure rates, from most reliable (top) to least reliable (bottom).
 - Complete abstinence (avoiding intercourse) is the only totally effective method of contraception.

The reliabilities of different forms of contraception



Contraception (1 of 6)

- Sterilization surgery prevents sperm from reaching an egg and is very reliable.
 - In a **vasectomy**, each vas deferens is cut, preventing sperm from reaching the urethra.
 - In a **tubal ligation**, a short section from each oviduct is removed, and often the remaining ends are tied off.
 - Sterilization is safe, meant to be permanent (although it can sometimes be surgically reversed), and generally free from side effects.

Vasectomy: 정관 절제 수술

Tubal ligation: 난관 결찰

Contraception (2 of 6)

- An **intrauterine device (IUD)** is a T-shaped device placed within the uterus by a healthcare provider.
 - IUDs are safe and highly effective at preventing pregnancy for up to 12 years, but can be safely removed at any time.

Contraception (3 of 6)

- Oral contraceptives, or **birth control pills**, come in several different forms that contain **synthetic estrogen and/or a synthetic progesterone**.
 - Steady intake of these hormones simulates **their constant levels during pregnancy**.
 - In response to **being “fooled” that pregnancy has occurred**, no eggs are released.
 - In addition to pills, various combinations of these hormones are also available as a shot (Depo-Provera), implant, or skin patch.

A contraceptive skin patch containing birth control hormones



Contraception (4 of 6)

- If used correctly, methods that physically block the union of sperm and egg can be quite effective.
 - **Condoms** are sheaths usually made of latex that fit over the penis or within the vagina.
 - **Diaphragms** are dome-shaped rubber caps that cover the cervix.
 - Barrier devices are more effective when used in combination with **spermicides**, sperm-killing chemicals; spermicides used alone are unreliable.

Contraception (5 of 6)

- Temporary abstinence, also called **natural family planning** or the **rhythm method**, depends on refraining from intercourse during the days around ovulation, when fertilization is most likely.
 - Because the length of the reproductive cycle can vary from month to month, and because sperm can **survive for 3–5 days** within the female reproductive tract, natural family planning is unreliable.
 - Withdrawal of the penis from the vagina before ejaculation is also ineffective, because sperm may exit the penis before climax.

Contraception (6 of 6)

- If unprotected intercourse has already occurred, high doses of some formulations of birth control pills can be used as emergency contraception, also called **morning after pills (MAPs)**.
 - If taken **within 3 days of intercourse**, morning-after pills are about **75% effective** at preventing pregnancy.
 - If pregnancy has occurred, the drug **RU-486 (mifepristone)** can induce an abortion, the termination of a pregnancy in progress.

Sexually Transmitted Infections (1 of 3)

- **Sexually transmitted infections (STIs)**, also called sexually transmitted diseases (STDs), are contagious diseases spread by sexual contact.
 - **Table 26.1** lists the STIs most common in the United States, organized by the type of infectious agent. Notice that bacteria, viruses, protists, and fungi can all cause STIs.
 - The most common bacterial STI is **chlamydia**. The primary symptoms are genital discharge and a burning sensation during urination. But chlamydia is frequently “silent”—half of infected men and three-quarters of infected women do not notice symptoms.

STIs Common in the United States

Table 26.1 STIs Common in the United States			
Disease	Microbial Agent	Major Symptoms and Effects	Treatment
Bacterial			
Chlamydia	Chlamydia trachomatis	Genital discharge, itching, and/or painful urination; often no symptoms in women; pelvic inflammatory disease (PID)	Antibiotics
Gonorrhea	Neisseria gonorrhoeae	Genital discharge; painful urination; sometimes no symptoms in women; PID	Antibiotics
Syphilis	Treponema pallidum	Ulcer (chancre) on genitalia in early stages; spreads throughout body and can be fatal if not treated	Antibiotics can cure in early stages
Viral			
Genital herpes	Herpes simplex virus type 2, occasionally type 1	Recurring symptoms: small blisters on genitalia, painful urination, skin inflammation; linked to cervical cancer, miscarriage, birth defects	Drugs can prevent recurrences and treat symptoms, but there is no cure
Genital warts	Human papillomaviruses (HPV), multiple types	Painless growths on genitalia; HPV can cause cervical cancer	Removal by freezing; can be prevented by HPV vaccine
AIDS and HIV infection	HIV	Destruction of the immune system; see Chapter 24	Combination of drugs can slow progression
Protistan			
Trichomoniasis	Trichomonas vaginalis	Vaginal irritation, itching, and discharge; usually no symptoms in men	Antiprotist drugs
Fungal			
Candidiasis (yeast infections)	Candida albicans	Vaginal irritation, itching, and discharge; frequently acquired nonsexually	Antifungal drugs

임질
매독

생식기
혹

Sexually Transmitted Infections (2 of 3)

- In contrast to bacterial STIs, **viral STIs are not curable**.
 - They can be controlled by medications, but their symptoms and the ability to infect others remain a possibility throughout a person's lifetime.
 - One in five Americans is infected with **genital herpes**, caused by the herpes simplex virus type 2, a variant of the virus that causes cold sores.
 - The **human papillomavirus (HPV)** is another sexually transmitted virus. A vaccine can help protect against genital warts and infection by the HPV strains that cause 70% of cervical cancers.

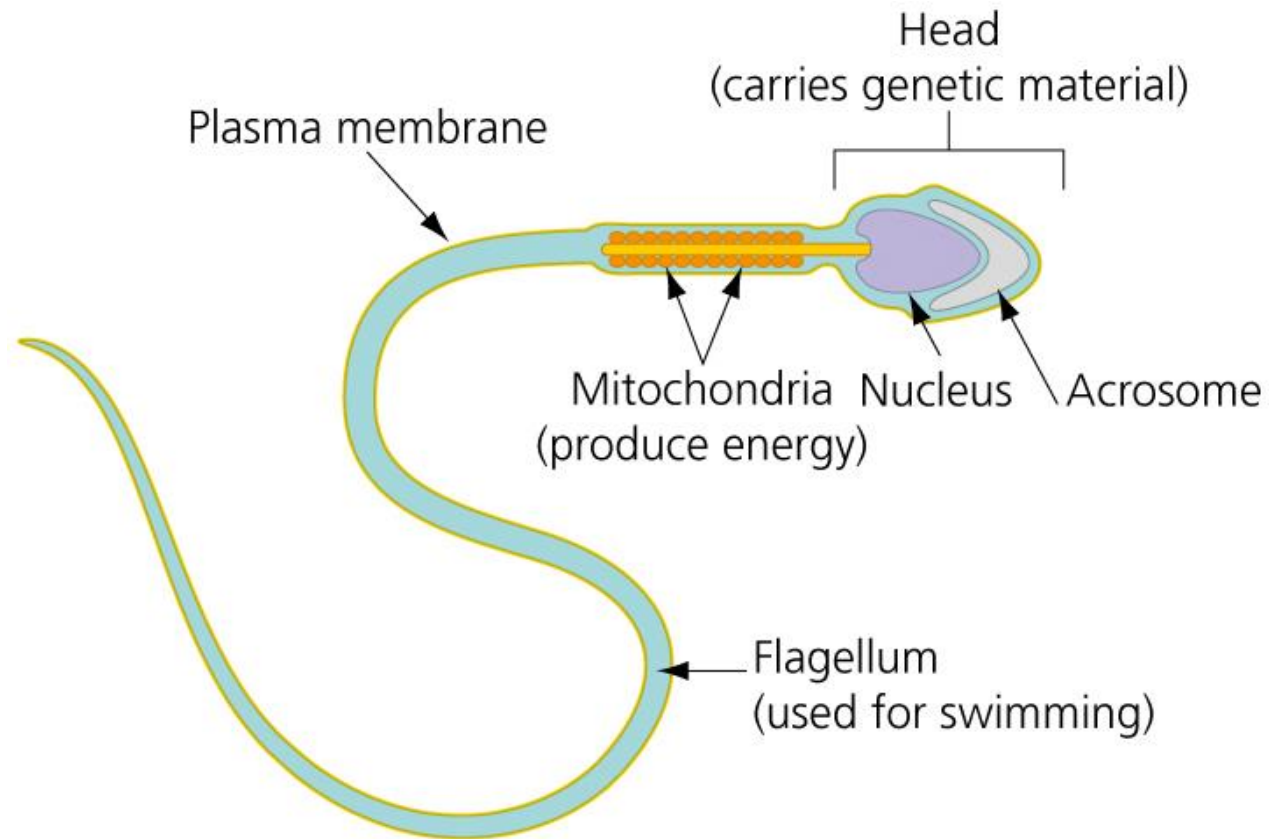
Sexually Transmitted Infections (3 of 3)

- **AIDS**, caused by HIV, poses one of the greatest health challenges in the world today, particularly among the developing nations of **Africa and Asia**.
- But even within the United States, there are 50,000 new infections each year, one-third of which result from heterosexual contact.
- Anyone who is sexually active should have yearly medical exams, STI screening, and seek immediate help if any suspicious symptoms appear—even if they are mild. The best way to avoid the spread of STIs is abstinence. Alternatively, latex condoms provide the best protection.

Human Development: Fertilization by Sperm

- Embryonic development starts with **fertilization**, or conception, the union of sperm and egg to form a zygote.
- Sexual intercourse releases hundreds of millions of sperm into the vagina, but **only a few thousand sperm survive the trip to the egg in the oviduct** and only a single one can enter and fertilize the egg.

A human sperm cell

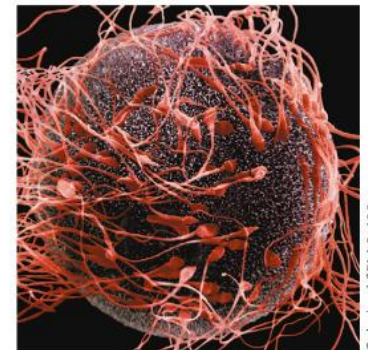
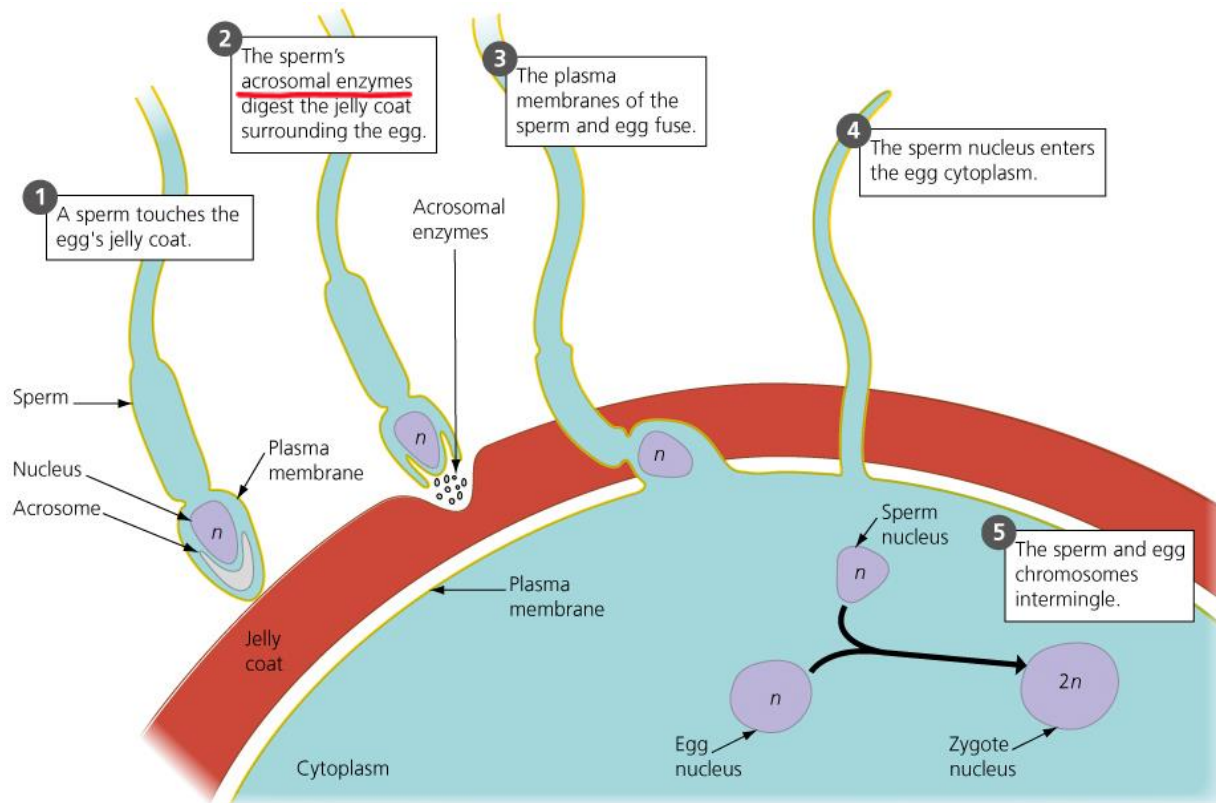


Fertilization by Sperm

- A mature human sperm has a streamlined shape that enables it to swim through fluids in the vagina, uterus, and oviduct.
- The sperm's head contains a haploid nucleus and is tipped with a membrane-enclosed sac called the acrosome.
- The middle of the sperm contains mitochondria that use high-energy nutrients from the semen, especially the sugar fructose, to fuel movement of the flagellum.
- **Figure 26.13** traces one sperm through the events of fertilization.

Acrosome: 첨체

Fertilization



Basic Concepts of Embryonic Development (1 of 4)

- The key to development in all organisms is that each stage occurs in a highly organized fashion.
 - Development begins with cleavage, a series of rapid cell divisions that results in a multicellular ball.
 - During cleavage, DNA replication, mitosis, and cytokinesis occur rapidly, but the total amount of cytoplasm remains unchanged.
 - As a result, the overall size of the embryo does not change; instead, each cell division partitions the embryo into twice as many smaller cells.
 - Cleavage continues as the embryo moves down the oviduct toward the uterus.

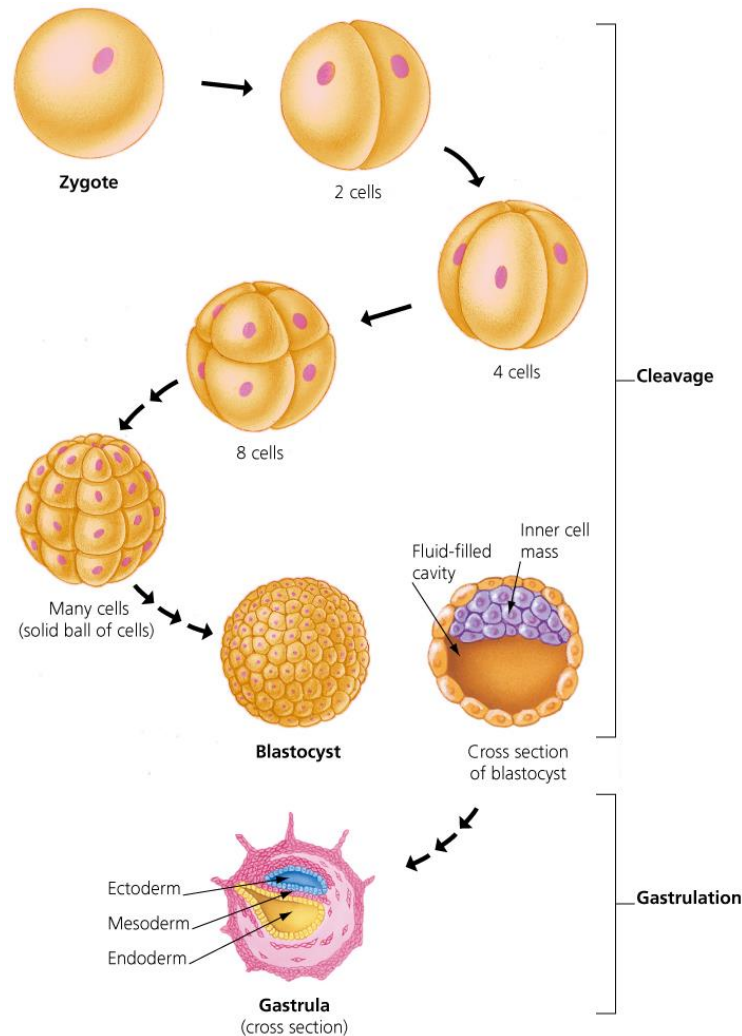
Basic Concepts of Embryonic Development (2 of 4)

- About 6–7 days after fertilization, the embryo has reached the uterus as a fluid-filled hollow ball of **about 100 cells called a blastocyst**.
- Protruding into the central cavity on one side of the human blastocyst is a small clump of cells called **the inner cell mass**, which will form the fetus.
- Not all embryos can complete development.
 - An **ectopic pregnancy** occurs when an embryo develops in the wrong location. This is invariably fatal to the embryo and can be dangerous to the mother, requiring immediate medical attention.

Basic Concepts of Embryonic Development (3 of 4)

- Many other embryos spontaneously **stop developing as a result of chromosomal or developmental abnormalities**. The result, a spontaneous abortion or miscarriage, occurs in as many as one-third of all pregnancies, often before the woman knows she is pregnant.
- About 9 days after conception, the cells of the early embryo begin an organized migration that produces the **gastrula**, with three main layers. At this stage, various cellular changes contribute to the formation of embryonic structures.

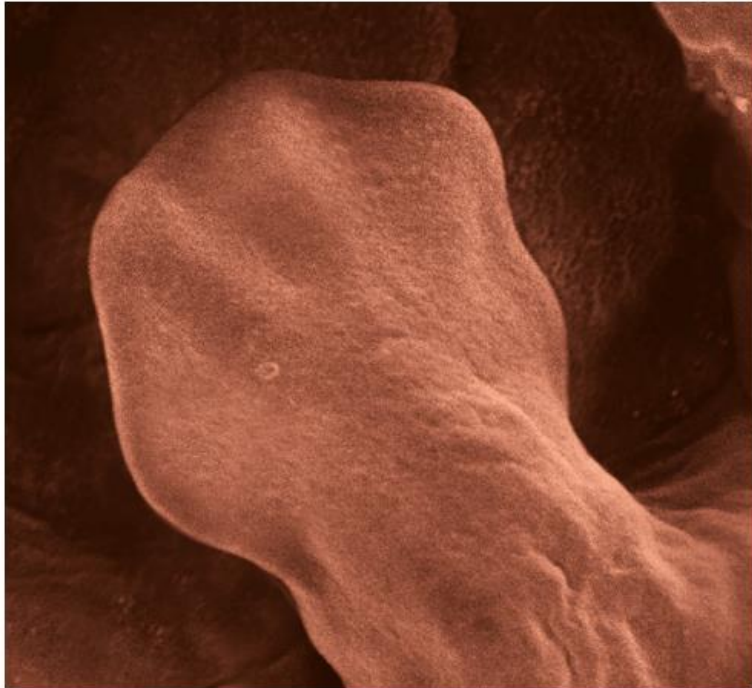
Early development of an embryo



Basic Concepts of Embryonic Development (4 of 4)

- All developmental processes depend on chemical signals passed between neighboring cells and cell layers, telling embryonic cells what to do and when.
 - Other signals might cause a cell to change shape or migrate from one location to another within the developing embryo.
 - Another key developmental process is programmed cell death, which kills selected cells, such as in the formation of fingers and toes.
 - Further along in development, a sequence of chemical signals between cells leads to greater specialization as organs begin to take shape.

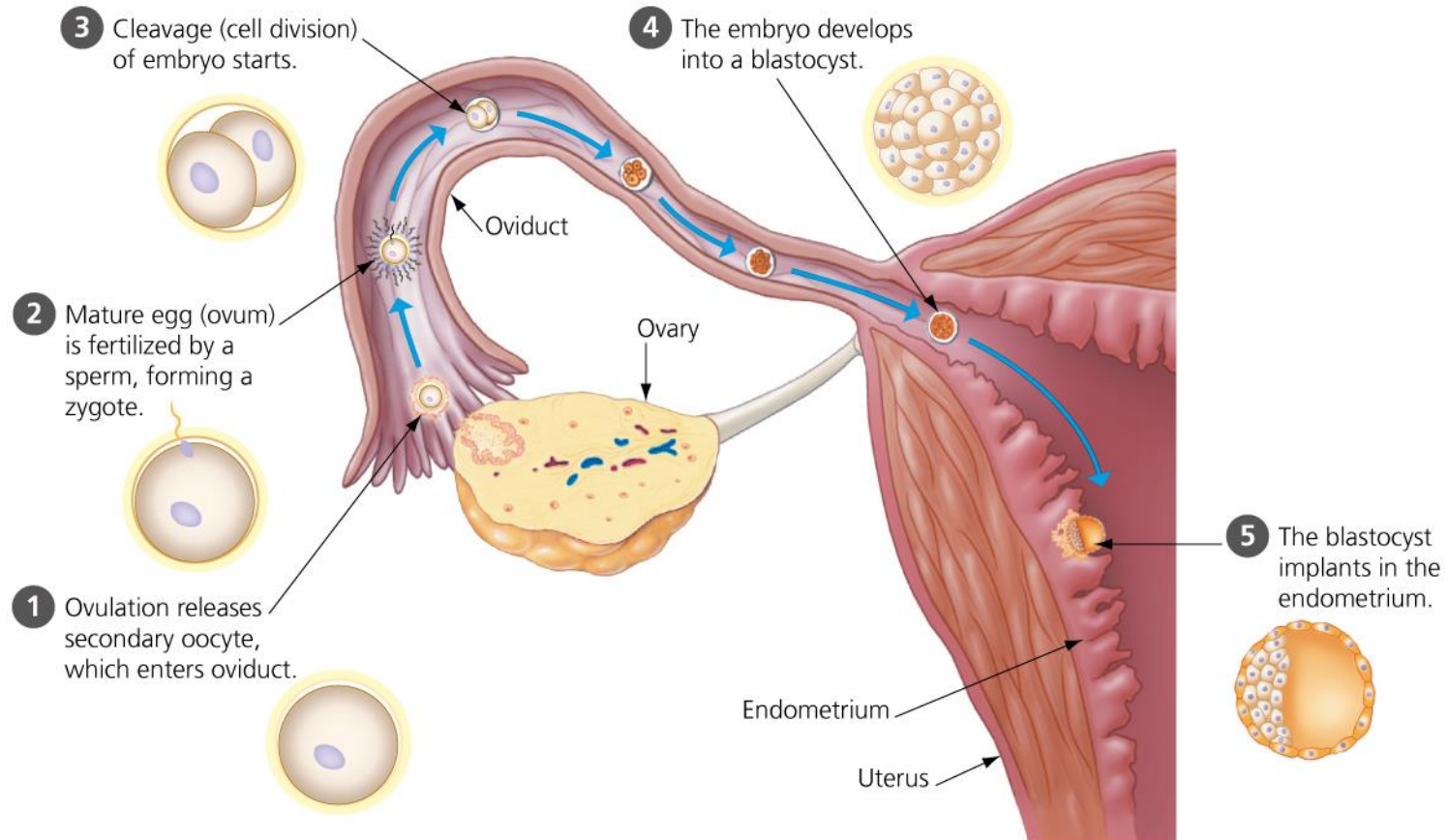
Programmed cell death in a developing human hand



Pregnancy and Early Development (1 of 4)

- **Pregnancy** is the carrying of developing young within the uterus.
 - In humans, pregnancy lasts about 266 days (38 weeks) from fertilization, but it is usually measured as 40 weeks (9 months) from the start of the last menstrual cycle.
 - The early stages in human development are summarized in **Figure 26.16**.

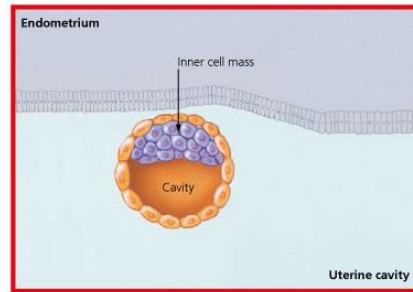
Early stages of human development



Pregnancy and Early Development (2 of 4)

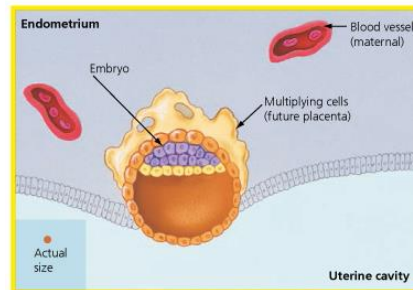
- By the time the embryo has become **a blastocyst**, it has a central fluid-filled cavity and inner cell mass.
 - These early cells are **stem cells**, with the potential to give rise to every type of cell in the body.
 - The outer cell layer becomes **part of the placenta**, the organ that provides nourishment and oxygen to the embryo and helps dispose of its wastes.
- **Figure 26.17** shows the embryo about a month after conception with several pieces of life-support equipment.

A human embryo: the first month

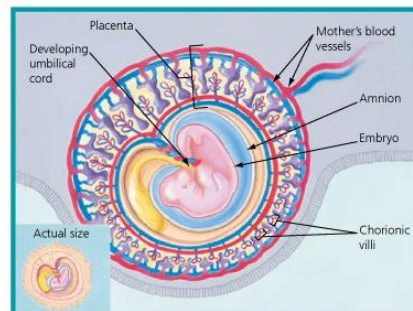


(a) **Day 6: Before implantation.** By 6 days after conception, the embryo has developed into a blastocyst.

January		
	0	Conception
Day 6		Day 10
		Day 31



(b) **Day 10: Implantation under way.** After implantation, the inner cell mass of embryonic stem cells will develop into the fetus. The outer layer of trophoblast cells will become the embryo's contribution to the placenta.



(c) **Day 31: The embryo and its life-support equipment**

Amnion: 양막

Pregnancy and Early Development (3 of 4)

- The **amnion** is a fluid-filled sac that encloses and protects the embryo.
- The **umbilical cord** serves as a lifeline between the embryo and the placenta. The placenta develops **chorionic villi** containing embryonic blood vessels closely associated with blood vessels of the mother's endometrium.
- The chorionic villi absorb nutrients and oxygen from the mother's blood by diffusion and pass these substances to the embryo. The villi also diffuse wastes from the embryo to the mother's bloodstream. The fetus has its own blood supply that does not mix with the mother's.

Umbilical cord: 탯줄

Placenta: 태반

Pregnancy and Early Development (4 of 4)

- The **placenta** provides other needs of the embryo.
 - It allows protective antibodies to pass from the mother to the fetus.
 - Unfortunately, chemicals in tobacco smoke, alcohol, and other drugs can also cross the placenta, and these harmful substances can cause developmental abnormalities and raise the chances of miscarriage.
 - Additionally, viruses—the German measles virus and HIV, for example—can cross the placenta and cause disease.

The Stages of Pregnancy: The First Trimester

- Pregnancy is divided into **three trimesters** of about 3 months each.
- The **First Trimester**
 - About 5 weeks after fertilization, the embryo has **developed a brain and spinal cord, four stumpy limb buds, a short tail, and primitive gill-like structures.**
 - Overall, a month-old human embryo looks pretty much like a month-old embryo of any other vertebrate species.

Trimester: 3개월
Stumpy: 뭉툭한

The development of a human

January						
		0				
		Conception				

February						

March						

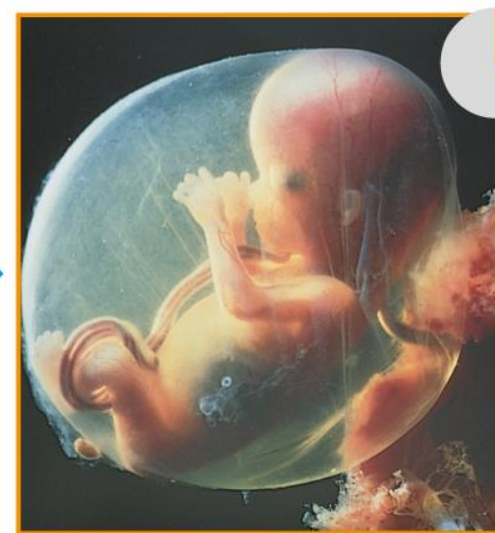
April						



5 weeks (35 days)



9 weeks (63 days)

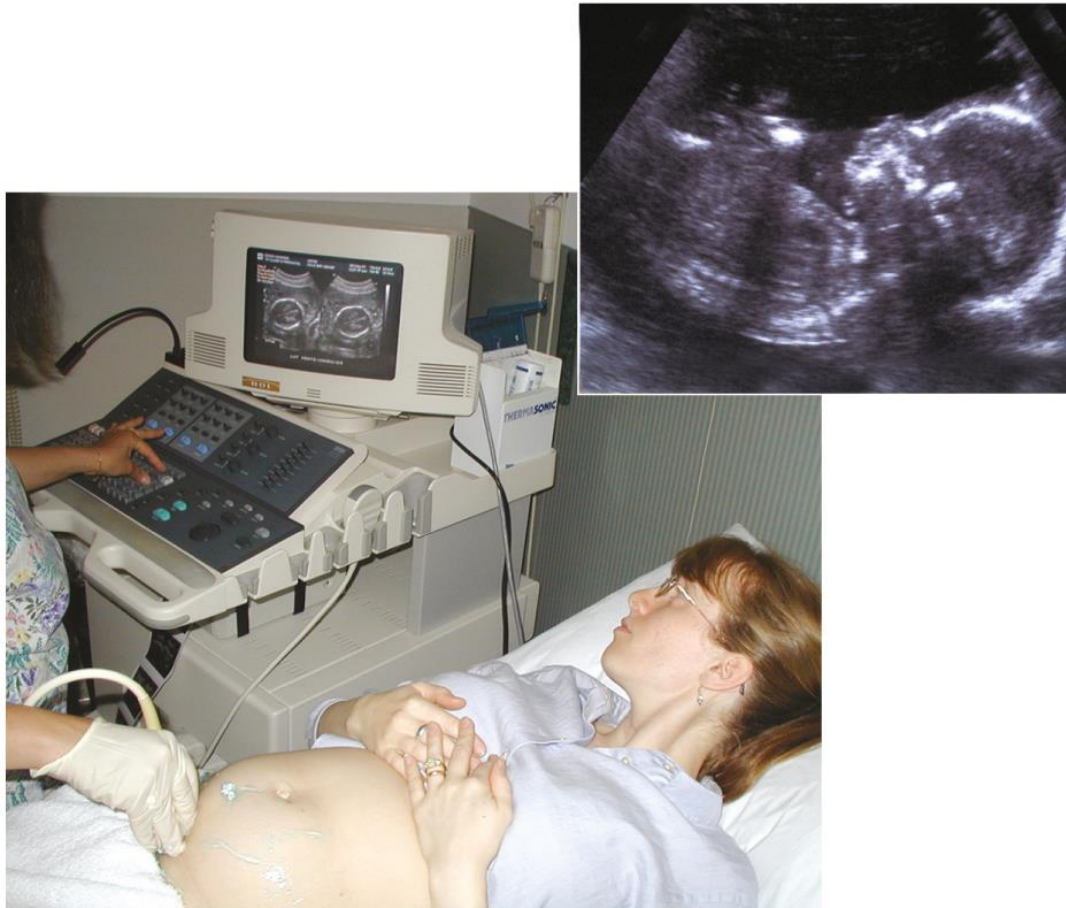


14 weeks (98 days)

The First Trimester

- About 9 weeks after fertilization, the embryo is called **a fetus**. It has a clear sac, the amnion, around it. It has all of its organs and major body parts, and the embryo's limb buds have become tiny arms and legs with fingers and toes.
- The sex of the fetus can be determined by an ultrasound exam.
- The main developmental changes during the second and third trimesters involve an increase in size and general refinement of the human features, not as dramatic as the changes of the first trimester.

Ultrasound imaging



The Second Trimester

- At 20 weeks, well into **the second trimester**, the fetus is about 19 cm (7.6 inches) long, weighs about half a kilogram (1 pound).
 - It has the face of an infant, complete with eyebrows and eyelashes. Its arms, legs, fingers, and toes have lengthened. It also has fingernails and toenails and is covered with fine hair.
 - The fetal heartbeat is detectable with a stethoscope and the mother can usually feel the fetus moving.
 - By the end of the second trimester, the fetus's eyes are open, its teeth are forming, and its bones have begun to harden.

Stethoscope: 청진기

The Third Trimester

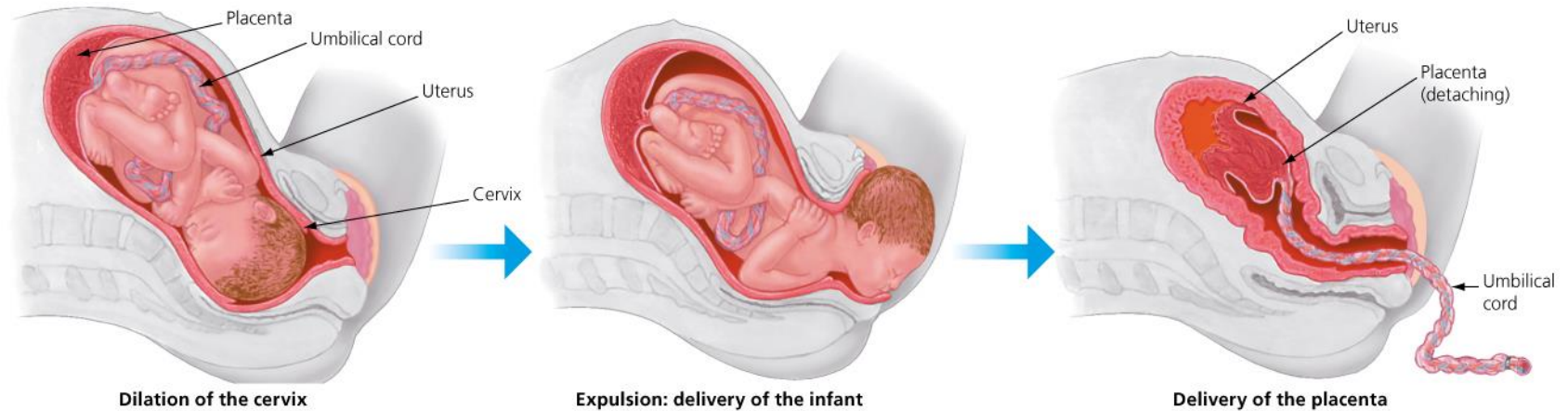
- The **third trimester** is a time of rapid growth as the fetus gains the strength it will need to survive outside the protective environment of the uterus.
 - Babies born prematurely—as early as 24 weeks—may survive, but they require special medical care after birth.
 - At birth, babies average about 50 cm (20 inches) in length and weigh 3–4 kg (6–8 pounds).

Childbirth (1 of 2)

- The birth of a child is brought about by **labor**, a series of strong, rhythmic uterine contractions.
- Several hormones bring about labor. One of the most important is **Oxytocin**, produced by the fetus's cells and, late in pregnancy, the mother's pituitary gland. It stimulates contractions of uterine muscles.
 - Uterine contractions stimulate the release of more oxytocin, which in turn increases the contractions—an example of **positive-feedback control**.
 - **Figure 26.20** shows the three stages of labor.

Labor: 분만

The three stages of labor



Childbirth (2 of 2)

- Hormones continue to be important after birth.
 - Decreasing levels of progesterone and estrogen allow the uterus to start returning to its pre-pregnancy state.
 - The pituitary hormone prolactin promotes milk production (called lactation) by mammary glands.
 - At first, a yellowish antibody-rich fluid called colostrum is secreted.
 - After 2–3 days, normal milk production begins.

Colostrum: 초유

Reproductive Technologies: Infertility

- Reproductive technologies can help solve problems related to the inability to conceive a child.
- **Infertility** is the inability to have children after one year of trying, usually due to problems in the man.
 - A man may not produce enough sperm or the sperm may not be vigorous enough to reach an egg. Some infertility is caused by **impotence** (erectile dysfunction), the inability to maintain an erection.
 - Female infertility can result from lack of eggs, failure to ovulate, or blocked oviducts. Other women can conceive but cannot support a growing embryo.

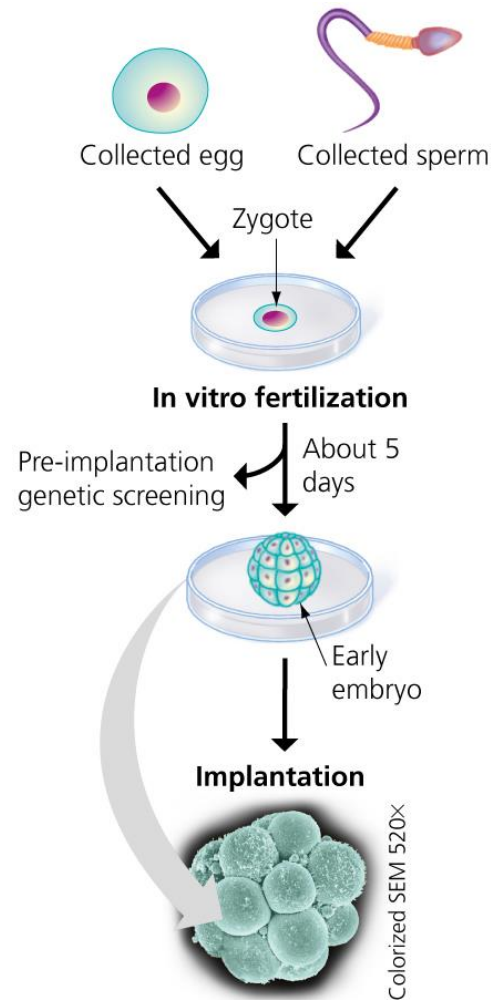
Infertility

- There are treatment options for infertility.
 - If a woman has normal eggs that are not being released properly, **hormone injections can induce ovulation**, but frequently **result in multiple pregnancies** (twins, triplets, or more).
 - If a woman has no eggs of her own, they can be **obtained from a donor** for fertilization and injection into the uterus.
 - If a woman cannot maintain pregnancy, she and her partner may enter into a legal contract with **a surrogate mother** who agrees to carry the couple's child to birth.

In Vitro Fertilization (1 of 2)

- **In vitro fertilization (IVF)** begins with the administration of drugs that promote the development of multiple eggs.
 - Eggs are surgically removed and then fertilized with sperm in a petri dish, allowed to develop for several days, and screened for genetic abnormalities.
 - **Blastocysts** that pass the screening are then **injected into a woman's uterus** at a time that coincides with her natural ovarian cycle.
 - Since its introduction in 1978, over 5 million babies have been born after in vitro fertilization treatments.

In vitro fertilization (2 of 2)



The Process of Science: Are Babies Conceived via In Vitro Fertilization as Healthy as Babies Conceived Naturally?

(1 of 2)

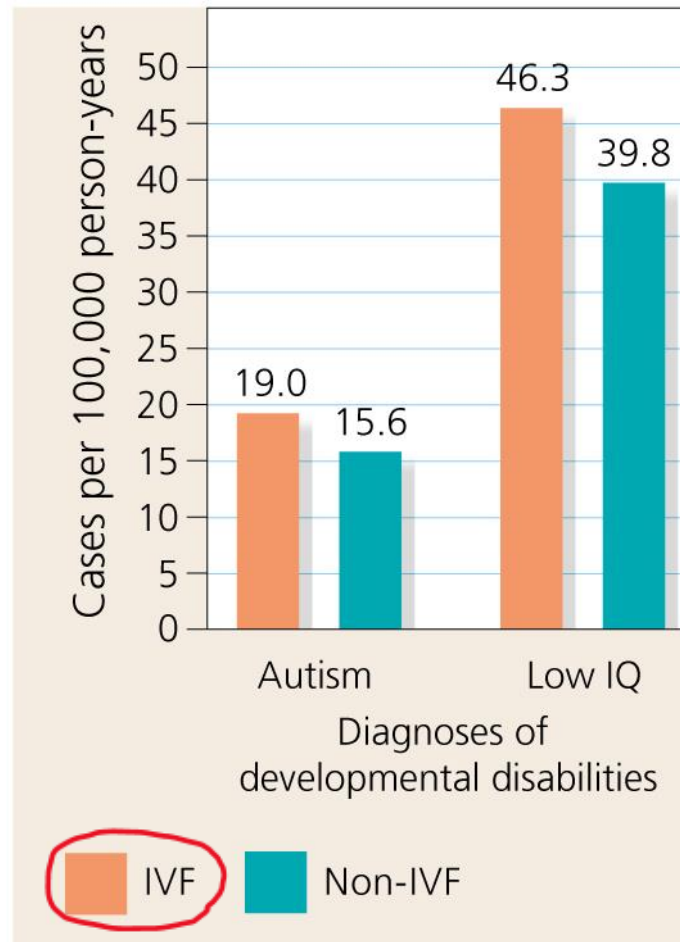
- **Background:** Worldwide, there are about 350,000 babies born each year after in vitro fertilization. Researchers in Sweden wondered whether developmental disabilities, such as autism and low IQ, were more prevalent or less prevalent in IVF children than in non-IVF children.
- **Method:** The researchers accessed a national database containing health data on over 2.5 million Swedish children born between 1982 and 2007. Of these, nearly 31,000 were born after IVF.

The Process of Science: Are Babies Conceived via In Vitro Fertilization as Healthy as Babies Conceived Naturally?

(2 of 2)

- The database also identified all children who had been diagnosed with autistic disorder or abnormally low IQ (below 70). By adjusting for age, the researchers could compare rates of developmental disabilities in the IVF and non-IVF populations.
- **Results** are summarized in **Figure 26.22**.
 - Only the difference in the rates for low IQ was statistically significant.
 - The difference in the rates of diagnosis of autism was not statistically significant.

Comparing IVF and non-IVF children



The Ethics of IVF (1 of 3)

- The preceding study is part of a growing body of evidence that IVF babies have a slightly greater risk of birth defects than babies conceived naturally.
- Despite such risks and the high cost (around \$10,000 for each attempt, whether it succeeds or not), IVF is now routinely performed at medical centers throughout the world and results in the birth of thousands of babies each year.

The Ethics of IVF (2 of 3)

- One major difference between natural and in vitro fertilization is that technology offers choices that nature does not.
 - Sperm can be sorted based on whether they contain an X or Y chromosome, increasing the likelihood of creating a zygote of a particular sex.
 - Cells can be harmlessly removed from early embryos and tested for disease-causing (or other) genes.
 - Thus, parents may have an increasing amount of information when choosing embryos to implant.

The Ethics of IVF (3 of 3)

- The concept of parenthood has been greatly complicated by modern reproductive technologies.
 - Mix-ups in fertility clinics have resulted in the implantation of embryos into the wrong women.
 - Divorcing couples have sued each other for custody of frozen embryos.
 - An infertile couple could purchase eggs from a woman, sperm from a man, and then hire a surrogate to carry the baby. Who, then, are the baby's true parents?
 - Each new reproductive technology raises moral and legal questions.

Evolution Connection: The “Grandmother Hypothesis”

- Menopause typically occurs after 500 menstrual cycles, usually around age 50, but in vitro fertilization has enabled some postmenopausal women to bear children up to age 70.
- How might menopause be adaptive?
- Menopause might increase a woman's fitness by allowing her to invest more time and energy in her children and grandchildren instead of having more children.

Human grandmothers often help care for their grandchildren



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